

CERD: Conditional Evidence Reconstruction and Decomposition for Incomplete Multimodal Learning

Shaowen Wan^a, Yanjun Lv^b, Lu Zhang^c, Dajiang Zhu^b, Bharat Biswal^a, Tianming Liu^d, Xiaobo Li^a and Lin Zhao^{a,*}

^aDepartment of Biomedical Engineering, New Jersey Institute of Technology, Newark, 07102, NJ, USA

^bDepartment of Computer Science and Engineering, The University of Texas at Arlington, Arlington, 76019, TX, USA

^cDepartment of Computer Science, Indiana University Indianapolis, Indianapolis, 46202, IN, USA

^dSchool of Computing, University of Georgia, Athens, 30602, GA, USA

ARTICLE INFO

Keywords:

Incomplete multimodal learning

Missing modality

Mixture-of-experts

Decision decomposition

Neurobiological prediction

ABSTRACT

Multimodal neurobiological prediction benefits from complementary imaging, genetic, clinical, and behavioral information, but real cohorts contain heterogeneous subject-specific missingness. Existing methods either operate only on the observed subset or reconstruct unavailable modalities, yet often do not distinguish generated representations from direct measurements when forming the final decision. We propose CERD, a Conditional Evidence Reconstruction and Decomposition framework for incomplete multimodal learning. CERD reconstructs missing-modality tokens from the same subject's observed modalities and uses provenance embeddings to preserve whether each representation is observed or generated. A Transformer with sparse top- k experts then models cross-modal interactions. Finally, global, modality-centered, and modality-pair-centered heads read the context-enriched slot features and produce complementary predictions. Their probabilities are combined according to learned input reliability, normalized-entropy confidence, and a branch prior. The stored branch probabilities and normalized weights exactly reproduce the neural prediction. Under arithmetic three-seed evaluation, CERD reaches 65.72% accuracy on three-class ADNI and 66.82% on a family-disjoint ABCD current-severity task. It leads accuracy, macro-F1, and macro-AUROC on both datasets among seven matched methods. Strict modality interventions further test whether label-free changes in the frozen model's output track the accuracy effect of removing each source.

1. Introduction

Neurobiological phenotypes are described at complementary levels by brain structure and function, genetic susceptibility, clinical measurements, and behavioral or environmental factors [12, 15, 35]. Multimodal learning can combine evidence absent from any single source through representation and fusion [57, 38, 43, 22, 1, 26, 30]. Its value in neurodevelopmental and neurodegenerative prediction, however, depends on using the evidence actually available for each subject rather than assuming a complete input set [4].

This assumption is frequently violated in research cohorts and clinical data. Modalities can be absent because of acquisition cost, protocol differences, contraindications, participant dropout, or failed quality control, producing many subject-specific availability patterns [50, 55, 11, 24, 56]. Restricting analysis to complete cases discards otherwise informative subjects, whereas replacing every missing modality with a population-level constant removes the dependence between the unavailable source and the subject's observed evidence. Robust incomplete-modality learning must consequently address two coupled questions: how to infer a task-relevant representation for an unavailable modality, and how strongly that inferred representation should influence the final prediction. Existing aggregation and alignment

methods avoid explicit completion but may leave recoverable cross-modal information unused [45, 21]. Generative approaches estimate missing observations or features, but many are specialized to commensurate imaging modalities or optimized primarily for reconstruction fidelity [49, 40, 3, 20, 59]. Flex-MoE supports arbitrary modality combinations through a missing-modality bank and sparse routing [54], yet its bank is indexed by the availability pattern and is therefore shared by subjects with the same observed subset. Less attention has been paid to jointly making completion subject-conditioned and making the downstream use of generated evidence reliability-aware.

The difficulty is not absence alone. Once a representation is inferred, the predictor must distinguish evidence supported by acquisition from evidence inferred through cross-modal association. Otherwise structural completion can be mistaken for evidential equivalence, allowing an uncertain reconstruction to influence fusion as if it were measured.

Missingness also changes what constitutes a useful explanation. A sparse router indicates where computation is allocated, but its gate value is not, by itself, the contribution of a modality to the prediction. Likewise, post-hoc attribution can audit a trained model, yet it need not reproduce the decision rule used during inference [23, 39, 31, 33]. For an incomplete subject, the most informative account distinguishes decision mass supported by observed representations from that supported by generated representations and shows how their certainty shapes the final output. This motivates a decision mechanism in which source status,

*Corresponding author.

 lin.zhao.1@njit.edu (L. Zhao)

ORCID(s):

predictive reliability, confidence, and branch contribution are represented explicitly and used in the prediction itself.

We propose CERD, a *Conditional Evidence Reconstruction and Decomposition* framework for incomplete multimodal learning. CERD maps heterogeneous modalities to a shared token space and reconstructs each unavailable target from a non-empty subset of the same subject's observed tokens. Provenance embeddings preserve whether a representation was observed or generated, and a Transformer with sparse top- k experts models interactions among the completed modality tokens. CERD then reads the fused state through global, modality-centered, and modality-pair-centered prediction heads and combines their probabilities using learned input reliability, normalized-entropy confidence, and a low-capacity prior. Because these branch predictions and weights are used directly in the forward pass, they provide an exact numerical decomposition of the neural probability.

We evaluate CERD on the Adolescent Brain Cognitive Development (ABCD) Study and the Alzheimer's Disease Neuroimaging Initiative (ADNI) [8, 47]. ABCD distinguishes low current symptom burden, subthreshold current ADHD, and current full ADHD through disjoint K-SADS symptom intervals and clinical conditions. ADNI distinguishes cognitively normal, mild cognitive impairment, and Alzheimer's disease. Each study uses four modality groups and the incomplete-input protocol in Section 4.1. We report accuracy, macro-F1, and macro-AUROC. The principal contributions are:

- We propose CERD, a unified incomplete multimodal framework combining recovery, provenance, sparse specialization, and reliability-based prediction.
- Target-specific generators infer an unavailable latent modality from the same subject's observed subset, rather than from the missingness pattern alone.
- Observed and generated provenance conditions self-attention and top- k expert routing, making specialization sensitive to evidence origin.
- Global, modality-centered, and pair-centered predictions form an exact normalized probability decomposition. Three-seed studies test its components and source relevance by strict removal.

2. Related Work

2.1. Incomplete Multimodal Representation

Incomplete multimodal representation learning determines what information reaches a predictor when the observed subset varies across subjects [50, 55]. Subset-native methods directly encode the available inputs. HeMIS aggregates latent features with permutation-invariant statistics [11]; SMIL applies Bayesian meta-learning under severe missingness [24]; and M3Care models relations between patients and modalities in incomplete healthcare records [56]. Alignment methods instead make partial views compatible

with complete ones. ShaSpec separates shared and source-specific factors [45], whereas MissModal combines geometric, distributional, and semantic constraints [21]. Missing-modality prompts adapt multimodal Transformers for visual recognition and affective analysis [16, 10]. SimMLM couples dynamic modality experts with an objective that relates predictions from richer and reduced views [19]. These methods preserve or align information from the current subset without constructing a target representation for every missing slot.

Conditional recovery explicitly supplies absent information. Multimodal variational autoencoders combine posteriors from individual modalities to infer a shared latent variable [49], and generalized multimodal evidence lower bounds extend training across observed subsets [40]. At the feature level, MMIN predicts unavailable input representations [58]; ActionMAE reconstructs randomly dropped sensing features [48]; and DiCMoR transfers observed distributions into a missing-modality space with normalizing flows [46]. Hetero-modal variational encoder-decoders jointly complete images and segment anatomy [3]. AGDiC recovers latent neuroimaging features with adaptive graphs and masked cross-attention [20], while ACADiff synthesizes missing MRI or PET volumes from images and metadata [59]. CERD targets heterogeneous sources. It generates one token slot per modality, supervised by held-out observed representations.

Pattern-conditioned completion occupies a useful middle ground. Flex-MoE learns a missing-modality bank indexed by the observed combination before sparse expert fusion [54]. CERD advances from pattern-level substitution to subject-level inference: each target generator queries the current subject's observed tokens, and provenance marks the resulting representation for downstream fusion and predictive-reliability estimation.

Subset-native models retain only measured information, whereas conditional recovery restores a source-indexed layout that supports downstream interactions. For heterogeneous sources, recovery in a shared task space also avoids requiring imaging, genetic, clinical, and behavioral variables to share an observation space. CERD therefore reconstructs target-specific latent tokens, preserves their generated origin, and estimates their predictive trust instead of equating marginal reconstruction similarity with subject-specific utility.

2.2. Sparse Expert Fusion

Once a multimodal representation has been assembled, fusion determines how interaction capacity is allocated across its tokens. Transformer self-attention models cross-token context [42], whereas sparse mixtures of experts (MoE) condition the feed-forward transformation by routing each token to a small expert subset [37, 17, 5]. Routing designs control both specialization and expert utilization. BASE layers formulate balanced token assignment explicitly [18], expert-choice routing reverses the assignment direction so that experts select tokens [60], and MegaBlocks

realizes dynamic expert computation with block-sparse operators [7].

Sparse experts have also been adapted to modality structure. V-MoE established scalable expert routing for visual tokens [32], and LIMoE showed that a shared expert pool can develop differentiated language–image processing [25]. MoE++ augments learned feed-forward experts with zero, copy, and constant transformations [13], while Flex-MoE specializes routing for arbitrary observed-modality combinations [54]. Together, these studies establish sparse routing as a mechanism for content-dependent representation specialization.

Incomplete inputs add a second routing signal beyond semantic content: whether a modality is measured, unavailable, or reconstructed. A fusion model can vary its token set with the observed subset, encode the availability pattern, or restore the missing positions before interaction. The last option preserves a stable source-indexed structure across subjects, but it also requires the router to recognize that measured and generated tokens carry different evidence status. CERD meets both requirements by routing a completed token layout after adding provenance, so expert specialization can depend jointly on content, source position, and observation status.

Interaction-structured experts address a complementary aspect of multimodal fusion. MMoE separates experts for unique, redundant, and synergistic information [53], and I²MoE adds weakly supervised interaction objectives and sample-specific reweighting [51]. CERD places sparse experts inside the provenance-aware completed-token backbone, so routing transforms observed and generated evidence in their shared context. A separate reliability–evidence mechanism then determines how much each resulting decision view contributes to the prediction.

2.3. Interpretability and Decision Decomposition

Clinical and neurobiological models are often examined with post-hoc feature attribution, although clinical usefulness depends on whether the explanation is understandable and actionable for its intended user [41, 34, 2]. SHAP, integrated gradients, Grad-CAM, LIME, and counterfactual explanations estimate input relevance under different reference, gradient, localization, surrogate, or intervention constructions [23, 39, 36, 31, 44]. Multimodal variants shift the unit of analysis from individual features to information sources: MM-SHAP quantifies modality contributions and can expose single-modality dominance [28], while Layer-Wise Modality Decomposition traces modality-specific contributions through a pretrained fusion network [29]. These methods are valuable for examining arbitrary architectures, but the resulting attribution is external to the original prediction rule and may change with the explanation protocol.

Interaction-based interpretation asks whether modalities contribute unique, redundant, or synergistic information. MMoE and I²MoE encode these categories in specialized experts [53, 51], and LSMI estimates sample-level interactions using pointwise information-theoretic quantities [52]. Such analyses characterize relationships among modalities,

but they do not automatically decompose the probability produced by a classifier. Under missingness, the origin and dependability of the evidence also become part of the decision. CERD addresses this requirement by predicting from several slot-centered views of one fused state and normalizing their contributions using learned modality reliability and predictive evidence. The resulting weights provide an exact numerical decomposition of the classifier probability and expose how the network combines its decision views.

CERD connects functions usually treated separately: subject-conditioned generation supplies missing latent evidence, provenance conditions interaction and routing, and reliability controls decision mass. Thus reconstruction does not erase source status, and the final mixture remains part of inference.

3. Methodology

3.1. Overview and Problem Formulation

CERD is an end-to-end framework for classifying subjects with incomplete multimodal data. Figure 1 presents the complete CERD pipeline from an incomplete input to its class probability, then divides it into three stages. First, conditional reconstruction infers unavailable tokens from the subject’s observed context (see Section 3.2). Second, provenance-aware sparse-expert fusion transforms the completed layout without erasing evidence origin (see Section 3.3). Finally, reliability-aware decomposition combines global, modality-centered, and pair-centered predictions (see Section 3.4). The sparse MoE is the shared backbone, and the normalized branch mixture is the prediction itself.

Throughout, n , m , and b index subjects, modalities, and decision branches, respectively; C is the number of classes. Let $\mathcal{M} = \{1, \dots, M\}$ be the modality set. The binary variable o_n^m records whether subject n ’s input $\mathbf{x}_n^m$ is observed. A modality-specific encoder maps each observed input into P tokens of dimension D :

$$\mathbf{Z}_n^m = E_m(\mathbf{x}_n^m) \in \mathbb{R}^{P \times D}. \quad (1)$$

Let $\mathcal{O}_n = \{m : o_n^m = 1\}$ and $\mathcal{U}_n = \mathcal{M} \setminus \mathcal{O}_n$ denote the observed and missing sets. At least one modality must be observed, and every $m \in \mathcal{U}_n$ is generated from $\mathcal{O}_n$. CERD models $p(y_n \mid \{\mathbf{x}_n^m : m \in \mathcal{O}_n\}, \mathbf{o}_n)$, conditioning the prediction jointly on observed evidence and source availability. The availability vector determines provenance throughout the network, and generated evidence receives a learned predictive-reliability score below the observed reference value.

3.2. Conditional Evidence Reconstruction

A fixed missing-modality token assigns the same surrogate representation to every subject with the same availability pattern, so it cannot reflect differences in their observed evidence. CERD instead reconstructs each missing target conditionally from the current subject’s available modalities. During training, a non-empty proper subset $\mathcal{S}_n \subsetneq \mathcal{O}_n$ is sampled as context and $\mathcal{T}_n = \mathcal{O}_n \setminus \mathcal{S}_n$ supplies reconstruction targets. Context modalities are independently dropped

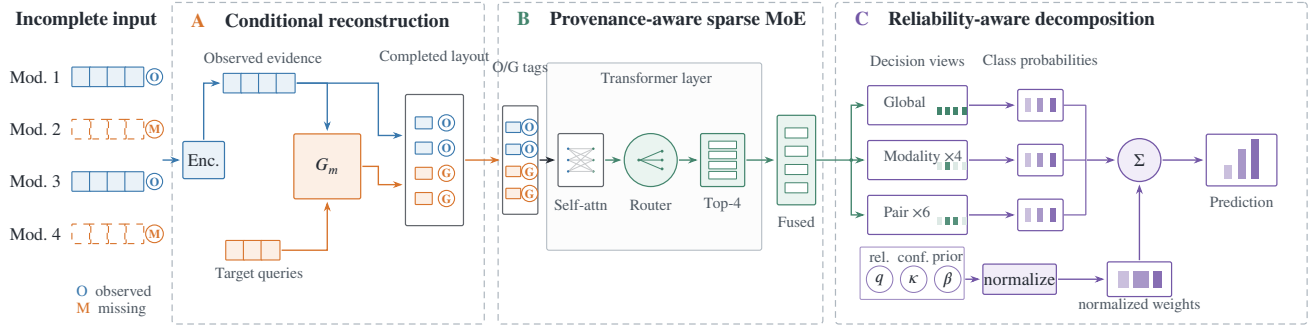


Figure 1: CERD reconstructs unavailable modality tokens from observed subject context, preserves observed/generated provenance through sparse expert fusion, and combines multigranular branch predictions using normalized reliability-aware weights.

and the draw is repaired when necessary so that both S_n and T_n remain non-empty. This stochastic observed-subset construction exposes each generator to both rich and sparse contexts rather than only leave-one-modality-out cases.

For a reconstruction target $m \in T_n$, we first collect the context tokens in Z_n^S . A target-specific generator then uses P learned queries Q_m to retrieve the evidence needed for modality m :

$$\begin{aligned} Z_n^S &= \text{Concat}_{j \in S_n} Z_n^j, \\ \hat{Z}_n^m &= G_m(Q_m, Z_n^S). \end{aligned} \quad (2)$$

The generator applies cross-attention followed by feed-forward refinement. Its refined query sequence U_n^m passes through a channel gate:

$$\hat{Z}_n^m = U_n^m \odot \sigma(W_m^g U_n^m + b_m^g). \quad (3)$$

The gate suppresses query responses unsupported by the current context. At inference, all observed modalities form Z_n^S , and generators are invoked only for missing targets. Thus, subjects with the same missing pattern can receive different reconstructions whenever their observed evidence differs.

Reconstruction is supervised at the pooled and token levels. Let $\bar{Z}$ denote token averaging, R_m a target-specific projector, LN token-wise layer normalization, and sg stop-gradient. For target (n, m) ,

$$\begin{aligned} \ell_{\text{pool}}^{n,m} &= 1 - \cos\left(R_m(\bar{Z}_n^m), \text{sg}[R_m(\bar{Z}_n^m)]\right), \\ \ell_{\text{tok}}^{n,m} &= \text{SmoothL1}\left(\text{LN}(\hat{Z}_n^m), \text{sg}[\text{LN}(Z_n^m)]\right). \end{aligned} \quad (4)$$

For a mini-batch $\mathcal{N}$, these target losses are averaged as

$$\begin{aligned} N_{\text{rec}} &= \sum_{n \in \mathcal{N}} |T_n|, \\ \mathcal{L}_{\text{rec}} &= \frac{1}{N_{\text{rec}}} \sum_{n \in \mathcal{N}} \sum_{m \in T_n} \left(\ell_{\text{pool}}^{n,m} + \lambda_{\text{tok}} \ell_{\text{tok}}^{n,m} \right). \end{aligned} \quad (5)$$

The pooled term preserves subject-level semantic direction, whereas the token term preserves the ordered slot structure.

Stop-gradient fixes the encoded target while the generator learns to approach it. One generator is reused for every context of target m : Q_m specifies the target, while Z_n^S specifies the subject and available evidence. CERD therefore avoids a separate generator for every missingness pattern.

The two reconstruction levels constrain complementary failure modes. Pooled alignment alone can preserve a subject-level mean while losing slot organization, whereas token alignment alone can reproduce local structure without retaining the target’s semantic direction. Their joint use makes generated tokens compatible with the fixed fusion layout at both resolutions.

3.3. Provenance-Aware Sparse MoE Fusion

Reconstruction supplies a complete latent layout, but reconstructed information should not be indistinguishable from an observed measurement. Before fusion, CERD therefore adds an observed or generated provenance embedding:

$$\tilde{Z}_n^m = \begin{cases} Z_n^m + \mathbf{e}^{\text{obs}}, & o_n^m = 1, \\ \hat{Z}_n^m + \mathbf{e}^{\text{gen}}, & o_n^m = 0. \end{cases} \quad (6)$$

These embeddings retain token origin while keeping all modalities in the same latent space.

The marked tokens are concatenated and processed by global self-attention. The Transformer feed-forward block is replaced by a sparse MoE layer. For token $\mathbf{h}$, router $\mathbf{g}$ defines the active expert set $\mathcal{A}(\mathbf{h})$, after which only those expert outputs are combined:

$$\begin{aligned} \mathcal{A}(\mathbf{h}) &= \text{TopK}(\mathbf{g}(\mathbf{h}), k), \\ \text{MoE}(\mathbf{h}) &= \sum_{e \in \mathcal{A}(\mathbf{h})} \pi_e(\mathbf{h}) F_e(\mathbf{h}). \end{aligned} \quad (7)$$

Here, F_e is expert e , and π_e is its softmax weight within $\mathcal{A}(\mathbf{h})$. Because provenance is added before routing, expert selection can respond to modality content and observed-generated status. A load-balancing term promotes utilization of the expert pool. The fused sequence is finally separated by modality and pooled into context-enriched features $\{\mathbf{f}_n^m\}_{m=1}^M$.

The attention and expert stages have complementary roles. Global self-attention communicates evidence across

every modality position, whereas sparse feed-forward experts specialize the transformation applied to each resulting token. The router therefore adapts computation to both the subject and the observed-generated composition without creating a separate network for every missingness pattern. With $M = 4$, the shared sequence contains 64 tokens on ADNI and 32 on ABCD. ADNI activates the top four of 16 experts per token, whereas ABCD activates the top two of four. The MoE remains a shared representation backbone, and all subsequent diagnostic branches reuse its pooled modality features.

Routing follows self-attention, so expert assignments are computed from tokens that already encode whole-subject context rather than isolated unimodal features. Sparse specialization therefore refines cross-modal evidence while preserving one common modality layout; it does not partition the sources into independent processing streams.

3.4. Reliability-Aware Evidence Decomposition

A single readout compresses the fused state into one decision view. CERD instead uses three readout granularities. Let 0 denote the global branch, $m \in \mathcal{M}$ a modality-centered branch, and $(i, j) \in \mathcal{P} = \{(i, j) : i < j\}$ a pair-centered branch. The complete branch set is $\mathcal{B} = \{0\} \cup \mathcal{M} \cup \mathcal{P}$, and every branch produces a class-probability vector $\mathbf{p}_n^b$. Because the readouts follow global attention and sparse experts, they are anchored to particular modality positions but retain cross-modal context. A pair branch receives

$$\mathbf{u}_n^{(i,j)} = [\mathbf{f}_n^i; \mathbf{f}_n^j; \mathbf{f}_n^i \odot \mathbf{f}_n^j; |\mathbf{f}_n^i - \mathbf{f}_n^j|], \quad (8)$$

which represents agreement and discrepancy between its two anchors. With four modalities, $\mathcal{B}$ contains one global, four modality-centered, and six pair-centered heads.

CERD then separates learned input reliability from predictive confidence. Let a_m be a small modality-specific reliability head. Because every missing target is generated from at least one observed source, modality reliability has two cases:

$$r_n^m = \begin{cases} 1, & o_n^m = 1, \\ \sigma(a_m(\mathbf{f}_n^m)), & o_n^m = 0. \end{cases} \quad (9)$$

Observed evidence establishes the reference value, while generated evidence receives a subject-dependent score in $(0, 1)$. Since r_n^m is optimized through the prediction objective, it quantifies task-specific predictive trust. The raw scores are estimated independently; normalization is applied when branches compete for decision weight and when cohort-level shares are visualized.

Branch reliability is then obtained directly from the modalities anchoring each readout:

$$q_n^b = \begin{cases} \frac{1}{M} \sum_{m=1}^M r_n^m, & b = 0, \\ r_n^m, & b = m, \\ \sqrt{r_n^i r_n^j}, & b = (i, j). \end{cases} \quad (10)$$

The global mean remains on the scale of one reliability score, while the geometric mean requires support from both pair anchors. Predictive confidence is measured independently from the entropy H of the C -class branch distribution:

$$\kappa_n^b = \max\left(1 - \frac{H(\text{sg}[\mathbf{p}_n^b])}{\log C}, \epsilon\right). \quad (11)$$

This score measures distributional concentration and is used to compare decision views [9, 14, 27].

Because every modality position is either observed or conditionally generated, all decision views are evaluated. Original availability enters through provenance and Eq. (9). A learned scalar β_b supplies a prior for branch b . The unnormalized evidence score, normalized branch weight, and final prediction are

$$\begin{aligned} s_n^b &= (q_n^b + \epsilon)(\kappa_n^b + \epsilon)e^{\beta_b}, \\ w_n^b &= \frac{s_n^b}{\sum_{b' \in \mathcal{B}} s_n^{b'}}, \\ \mathbf{p}_n &= \sum_{b \in \mathcal{B}} w_n^b \mathbf{p}_n^b. \end{aligned} \quad (12)$$

Thus, reliability q_n^b , confidence κ_n^b , and prior β_b have separate roles before normalization. The stored $\mathbf{p}_n^b$ and w_n^b are forward-pass quantities, and their weighted sum exactly reproduces $\mathbf{p}_n$.

Reliability and confidence serve distinct roles. A generated branch may be sharply peaked yet weakly supported by its inferred input; an observed view may be reliable but class-ambiguous. Their product requires both source support and discriminative concentration, while the branch prior captures stable utility without replacing subject-dependent variation.

For visualization, branch weights are summarized at the modality level. A modality-centered weight is assigned to its anchor, a modality-pair-centered weight is divided equally between its two anchors, and the global weight is distributed uniformly:

$$d_n^m = w_n^m + \frac{1}{2} \sum_{j \neq m} w_n^{(m,j)} + \frac{w_n^0}{M}. \quad (13)$$

The resulting modality decision allocation is a descriptive summary of the model decision, rather than an additional prediction mechanism.

3.5. Training Objective

CERD is trained end to end with five losses, each attached to a defined part of the method chain:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_b \mathcal{L}_{\text{branch}} + \lambda_r \mathcal{L}_{\text{rec}} + \lambda_g \mathcal{L}_{\text{balance}} + \mathcal{L}_{\text{rob}}. \quad (14)$$

Here, $\mathcal{L}_{\text{cls}} = |\mathcal{N}|^{-1} \sum_{n \in \mathcal{N}} \text{CE}(y_n, \mathbf{p}_n)$ supervises the fused probability, and $\mathcal{L}_{\text{rec}}$ is defined in Eq. (5). Each branch is also trained directly. We first normalize its stopped reliability

within mini-batch $\mathcal{N}$, then apply reliability-weighted cross-entropy:

$$\alpha_n^b = \frac{\text{sg}[q_n^b]}{\sum_{u \in \mathcal{N}} \text{sg}[q_u^b] + \epsilon}, \quad (15)$$

$$\mathcal{L}_{\text{branch}} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \sum_{n \in \mathcal{N}} \alpha_n^b \text{CE}(y_n, \mathbf{p}_n^b).$$

This term supervises every readout independently of its current mixture weight. For sparse layer ℓ , let $\mathbf{I}_\ell$ collect the router probability received by each expert and $\mathbf{L}_\ell$ its assigned-token counts. The load loss $\mathcal{L}_{\text{balance}} = \sum_\ell [\text{CV}^2(\mathbf{I}_\ell) + \text{CV}^2(\mathbf{L}_\ell)]$ promotes balanced expert use.

Robustness is trained from a strict non-empty reduced view $\tilde{\mathcal{O}}_n \subset \mathcal{O}_n$. Let $\mathbf{p}_n$ and $\tilde{\mathbf{p}}_n$ be the original- and reduced-view predictions. The batch-averaged robustness loss is

$$\mathcal{L}_{\text{rob}} = \frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \left[\lambda_d \text{CE}(y_n, \tilde{\mathbf{p}}_n) + \lambda_{\text{sd}} T^2 \text{KL}(\text{sg}[\mathbf{p}_n^{(T)}] \parallel \tilde{\mathbf{p}}_n^{(T)}) + \lambda_{\text{mf}} [\text{CE}(y_n, \mathbf{p}_n) - \text{CE}(y_n, \tilde{\mathbf{p}}_n)]_+ \right]. \quad (16)$$

Superscript (T) denotes temperature-scaled probabilities. The three lines respectively train reduced-view classification, transfer the original-view class distribution, and preserve the advantage of richer evidence. Reduced-view training constrains the final decision across availability patterns, whereas stochastic reconstruction contexts provide target supervision to the generators.

These two subset constructions impose different invariances. Reconstruction holds out observed targets to learn cross-modal recovery, whereas reduced-view training perturbs availability at the prediction level and constrains the resulting class distribution. Their combination couples target recovery to stable decisions rather than optimizing reconstruction and robustness as separate objectives.

3.6. Inference and Computational Structure

At inference, CERD first encodes the observed sources and conditionally generates only the missing targets. All modality positions then enter the provenance-aware backbone in a fixed MP -token layout. Sparse routing activates k feed-forward experts per token, while the attention path and parameters remain shared across availability patterns. The branch heads add only $1 + M + \binom{M}{2}$ shallow classifiers. Equation (12) then returns both the class probability and the exact weights that produced it. No reconstruction loss, reduced-view pass, or auxiliary explanation model is required during inference.

This organization has two practical consequences. First, the number of possible missingness patterns grows exponentially with M , but CERD does not store a pattern-specific completion vector or prediction network. Target-specific generators are reused with any non-empty context, and sparse experts specialize at the token level. Second, the decision trace is available in the ordinary forward output:

modality reliability describes the status of each supplied representation, branch probabilities describe the decisions supported at different granularities, and normalized weights state how those decisions were combined.

4. Experiments

4.1. Datasets and Preprocessing

ABCD. The ABCD Study characterizes adolescent brain, cognitive, behavioral, and health development [8]. We derive an ordered cross-sectional current-severity endpoint from the baseline parent K-SADS assessment. Let s be the larger positive-symptom count across the nine-item inattention and hyperactivity/impulsivity domains. Class 0 requires $s \in [0, 2]$, no current, past, partial-remission, or unspecified ADHD status, and no current functional impairment. Class 1 requires $s \in [3, 5]$, no current full ADHD, and a positive K-SADS onset/cross-setting condition. Class 2 requires $s \in [6, 9]$, current full ADHD, current impairment, and the onset/cross-setting condition. The non-overlapping symptom intervals make the middle group a current subthreshold state rather than a diagnosis-symptom discordance group; the endpoint describes severity, not longitudinal progression. The 3,000-participant cohort contains 1,733, 404, and 863 subjects in classes 0, 1, and 2. Family-disjoint training, validation, and test splits contain 2,134, 438, and 428 participants, with respective class counts (1,222, 277, 635), (262, 66, 110), and (249, 61, 118).

ABCD predictors form four model modalities. Imaging (785 variables) concatenates rs-fMRI network connectivity and regional variance, regional beta contrasts from SST, n-back, and MID task-fMRI, regional T1-weighted gray-matter intensity, and DTI-derived FA. Genetics comprises 1,172 directly encoded SNP dosages retained after training-set missingness, minor-allele-frequency, and LD pruning within 14 prespecified gene windows; no PCA, PRS, or ancestry components are used. The cognition and health group contains 90 NIH Toolbox, WISC, Little Man, RAVLT, sleep, and physical-activity variables. The behavior and environment group begins with 221 non-ADHD behavioral, family, neighborhood, socioeconomic, and address-linked exposure variables, of which 219 survive training-only screening. Target-defining K-SADS variables and CBCL ADHD/attention scores are excluded from every predictor group.

ADNI. The Alzheimer's Disease Neuroimaging Initiative (ADNI) is a multicenter longitudinal study of biomarkers for cognitive decline and Alzheimer's disease [47]. We classify participants as cognitively normal, mild cognitive impairment, or Alzheimer's disease. Its four modalities are 187 regional FreeSurfer structural-MRI measurements, 387,639 directly encoded SNPs, 4,189 clinical and cognitive variables, and 151 biospecimen measurements. The cohort contains 2,116 participants: 1,480 for training, 318 for validation, and 318 for testing. Their class counts are 648/522/310, 139/112/67, and 139/112/67, respectively. In both datasets, a subject is retained when at least one selected modality is available.

Missing-modality patterns. ADNI follows source-table availability: 27.64%, 28.62%, and 31.13% of the training, validation, and test participants are incomplete, respectively. Every incomplete ADNI participant lacks exactly one of the four modalities, and none is all-missing. For ABCD, a fixed label-independent mask augments source-table availability to approximately 15% incomplete participants per split. Among additionally masked complete cases, one, two, or three modalities are withheld in proportions 0.72, 0.23, and 0.05. The all-missing case is forbidden, and every method receives the same split-specific inputs. Table 1 gives the exact final counts.

Table 1: Incomplete participants in the fixed splits; entries are n/N (%).

Split	ADNI, n/N (%)	ABCD, n/N (%)
Training	409/1480 (27.64)	320/2134 (15.00)
Validation	91/318 (28.62)	66/438 (15.07)
Testing	99/318 (31.13)	64/428 (14.95)

Preprocessing. For ABCD, imaging and tabulated variables with excessive missingness or near-zero variance are removed, after which available entries are median-imputed and standardized using training statistics. Genetic selection is likewise frozen from training participants: SNPs exceeding 2% missingness or below 1% minor-allele frequency are removed, followed by LD pruning with a 50-SNP window, five-SNP step, and $r^2 = 0.8$ threshold. The retained 0/1/2 dosages are median-imputed and standardized without PCA. For ADNI, modality tables are joined by PTID and repeated FreeSurfer exports are deduplicated by database update stamp. Regional structural measurements are mean-imputed and standardized, genotypes are mode-imputed and transformed to $[-1, 1]$, and missing entries in clinical and biospecimen tables are filled by training-column means. In both cohorts, entry-level imputation is applied only within an available modality. A wholly unavailable modality remains masked and is reconstructed only by CERD. All preprocessing statistics and feature-selection decisions are estimated from the training split.

4.2. Experimental Setup

Implementation details.

CERD uses hidden dimension $D = 128$ and one four-head fusion layer. ADNI uses $P = 16$ tokens per modality with 16 experts and top-4 routing; ABCD uses $P = 8$, four experts, and top-2 routing. Each target generator contains two cross-attention layers. We use $(\lambda_b, \lambda_g) = (0.1, 0.01)$, $(\lambda_d, \lambda_{sd}, \lambda_{mf}) = (0.1, 0.15, 0.1)$, $T = 2$, and $\epsilon = 10^{-3}$. The reconstruction coefficient is 1.0 on ADNI and 0.5 on ABCD. ABCD models complete 100 epochs with AdamW and a learning rate of 1.0×10^{-4} ; the class weight for class c is proportional to $n_c^{-0.75}$. Every method uses the same rule. Checkpoints are selected by validation macro-F1 and the test set is evaluated once after selection. Results are arithmetic mean $\pm$ sample standard deviation over seeds

Table 2

Comparison on ADNI (%). AUC denotes macro-AUROC; the best values are boldfaced.

Method	Acc.	Macro-F1	AUC
Flex-MoE [54]	62.26 $\pm$ 1.09	60.62 $\pm$ 2.12	77.75 $\pm$ 1.03
I ² MoE [51]	64.05 $\pm$ 2.27	61.83 $\pm$ 1.89	79.35 $\pm$ 1.63
MoE++-corrected [13]	59.75 $\pm$ 0.54	58.85 $\pm$ 2.06	79.23 $\pm$ 1.39
AnyMod [6]	64.78 $\pm$ 4.63	63.55 $\pm$ 4.28	79.77 $\pm$ 2.12
AGDiC [20]	58.60 $\pm$ 3.36	56.54 $\pm$ 2.38	75.10 $\pm$ 1.36
ACADiff [59]	55.87 $\pm$ 0.48	52.14 $\pm$ 1.20	71.10 $\pm$ 1.24
CERD (Ours)	65.72$\pm$1.09	64.56$\pm$2.09	81.07$\pm$0.55

0/1/2 for ADNI and 31/32/33 for ABCD; probabilities are never ensembled across seeds.

Evaluation metrics. We report accuracy, macro-F1, and class-averaged AUROC. Accuracy measures overall decisions, macro-F1 weights every endpoint class equally, and AUROC summarizes class ranking independently of one operating point. The experiments ask whether CERD improves prediction under heterogeneous missingness, whether each stage contributes, whether performance is retained for incomplete participants, and whether model-output relevance agrees with controlled modality interventions. ABCD comparisons use the same family-disjoint split and fixed masks for every method, with training-split preprocessing throughout.

The incomplete stratum is reported separately because aggregate performance can conceal a model that succeeds mainly on complete participants. Strict removal then holds participant identity fixed and turns the descriptive allocation into a testable prediction: removing evidence assigned greater decision mass should produce the larger score change.

4.3. Comparison with Baselines

Compared methods. We compare CERD with six representative methods for incomplete multimodal learning. Flex-MoE combines a missing-modality bank with sparse routing [54]. I²MoE models unique, redundant, and synergistic multimodal interactions [51], whereas MoE++-corrected introduces learned, zero, copy, and constant experts [13]. AnyMod uses modality queries and a task anchor to accommodate arbitrary input combinations [6]. AGDiC and ACADiff are evaluated through task-matched feature-space implementations because their original formulations operate on different neuroimaging representations [20, 59]. All methods use the same participant splits, modality groups, missingness patterns, and evaluation metrics.

Main results. Table 2 shows that CERD ranks first on all three ADNI metrics. Relative to the strongest competing entry in each column, its mean accuracy, macro-F1, and AUROC are higher by 0.94, 1.01, and 1.30 points, respectively. The concurrent gains in macro-F1 and AUROC indicate that the benefit is not confined to the majority diagnosis: CERD improves class-balanced decisions while preserving stronger one-vs-rest ranking.

Table 3

Comparison on family-disjoint, three-class ABCD current-severity classification with approximately 15% incomplete participants (%). AUC denotes macro-AUROC; best means are boldfaced.

Method	Acc.	Macro-F1	AUC
Flex-MoE	63.79±2.60	51.44±2.28	74.69±1.30
I ² MoE	63.24±0.82	49.46±1.27	74.60±0.92
MoE++-corrected	64.64±2.22	52.06±4.35	74.44±1.81
AnyMod	62.15±2.00	49.09±0.61	69.17±4.68
AGDiC	62.69±3.04	48.83±0.84	71.71±2.66
ACADiff	63.16±1.18	49.05±1.56	70.90±0.78
CERD (Ours)	66.82±2.14	55.18±1.16	77.90±0.55

Table 4

CERD performance across test availability strata (%), mean ± SD).

Data	Input	<i>n</i>	Acc.	Macro-F1	AUC
ADNI	Complete	219	65.45±1.47	64.65±1.44	80.99±0.87
	Incomplete	99	66.33±2.92	62.86±4.08	82.35±1.30
ABCD	Complete	364	67.22±2.22	54.83±2.11	77.51±1.33
	Incomplete	64	64.58±4.77	55.06±4.59	80.13±3.81

On ABCD, CERD gives the highest accuracy, macro-F1, and macro-AUROC, exceeding the strongest competing means by 2.18, 3.12, and 3.20 points, respectively. The larger macro-F1 margin indicates that the gain is not explained by the low-symptom majority alone; the simultaneous AUC increase shows stronger one-vs-rest ranking across the ordered severity groups.

Table 4 shows that CERD remains operational for incomplete participants in both cohorts. On ABCD, the complete and incomplete strata have similar mean metrics, while the 64-subject incomplete stratum has much larger seed variability. Because these strata contain different participants, they describe observed deployment conditions rather than the causal effect of removing a source; the matched within-participant interventions in Section 4.5 address the latter question.

4.4. Ablation Study

Compared variants. Each control starts from the dataset’s full configuration and removes one named mechanism. Conditional completion is bypassed; provenance embeddings are suppressed; expert diversity is removed by setting $E = k = 1$; or multigranular decomposition is reduced to the global head with its specialist objectives removed. For ABCD, a capacity-aligned Dense FFN replaces sparse expert routing, and the reliability control fixes every usable r_n^m to one while retaining entropy confidence and learned branch priors. Every variant is independently trained and validation-selected with the same three seeds as its main result. The two datasets are reported separately because their missingness mechanisms and endpoints differ.

Ablation results. Tables 5 and 6 identify complementary roles rather than interchangeable additions. Reducing CERD to the global head causes the largest accuracy loss on both datasets: 3.25 points on ADNI and 3.19 on ABCD.

Table 5

Component analysis on the full ADNI test cohort (%), mean ± SD). Best means are boldfaced.

Configuration	Acc.	Macro-F1	AUC
w/o Conditional Completion	63.73±0.79	62.82±1.58	79.48±0.41
w/o Provenance Embeddings	63.52±1.57	63.41±1.95	80.50±0.70
w/o Expert Diversity ($E = 1$)	64.05±0.65	63.70±0.59	81.13±1.02
w/o Multigranular Decomposition	62.47±0.79	62.36±1.00	79.72±1.32
w/o Reliability-aware Weights	64.05±1.49	63.88±0.21	80.95±0.19
Full CERD	65.72±1.09	64.56±2.09	81.07±0.55

Table 6

Component analysis on the full ABCD test cohort (%), mean ± SD). Best means are boldfaced.

Configuration	Acc.	Macro-F1	AUC
w/o Conditional Completion	64.41±2.77	53.34±2.43	75.37±3.08
w/o Provenance Embeddings	65.03±2.31	53.07±1.09	77.14±1.75
w/o Expert Diversity ($E = 1$)	64.95±1.76	52.82±2.22	77.79±0.47
Dense FFN	66.43±2.35	53.81±0.77	77.30±0.79
w/o Multigranular Decomposition	63.63±1.99	53.73±0.65	76.73±0.63
w/o Modality Reliability	66.04±3.28	53.91±1.88	77.44±1.38
Full CERD	66.82±2.14	55.18±1.16	77.90±0.55

The source-centered and pair-centered readouts therefore recover decision structure that one pooled representation does not preserve. Completion has the clearest availability-specific effect. Among the 64 incomplete ABCD participants, removing it lowers accuracy, macro-F1, and AUC from 64.58/55.06/80.13 to 58.33/45.87/75.08, linking reconstructed tokens directly to decisions under missing input rather than to reconstruction similarity alone. Removing provenance or modality reliability lowers overall ABCD macro-F1 by 2.10 and 1.27 points, respectively.

The expert controls separate sparse specialization from the remaining CERD mechanisms. Relative to $E = 1$, the full mixture gains 1.67/0.86 Accuracy/macro-F1 points on ADNI and 1.87/2.36 points on ABCD. On ABCD, full CERD also exceeds the capacity-aligned Dense FFN on all three metrics. The difference is most visible when a source is absent: CERD leads Dense by 1.56 Accuracy, 4.13 macro-F1, and 2.66 AUC points in the incomplete stratum. Sparse routing thus contributes through availability-dependent specialization, not merely additional parameters.

4.5. Modality Contribution Analysis

We characterize source use by strict leave-one-modality-out interventions on originally complete test participants. The selected input block is zeroed, marked unavailable, and excluded from conditional completion, holding participant identity and the trained model fixed. Model-output relevance is the rate at which this intervention changes the predicted class, normalized across the four modalities within each seed. Independently, positive accuracy decreases are normalized in the same way. Each four-entry vector therefore sums to 100%, and their correspondence tests whether label-free prediction sensitivity recovers the empirical performance ordering. All seed-level quantities are computed before arithmetic aggregation.

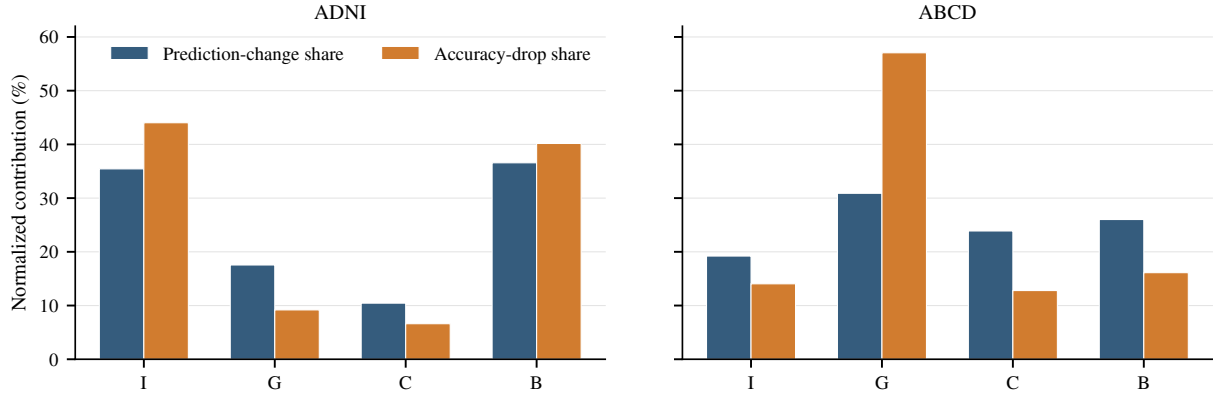


Figure 2: Normalized prediction-change and accuracy-drop shares under strict modality removal (three-seed means). I/G/C/B denote MRI/genetics/clinical/biospecimen on ADNI and imaging/genetics/cognition–health/behavior–environment on ABCD.

Table 7

Strict-removal modality audit (mean $\pm$ SD). Rel. is normalized prediction-change relevance; Drop is normalized positive accuracy decrease; Δ Acc. is the unnormalized decrease in percentage points.

Data	Modality	Rel. (%)	Drop (%)	Δ Acc.
ADNI	MRI	35.44 $\pm$ 3.79	44.03 $\pm$ 3.10	43.07 $\pm$ 1.47
	Genetics	17.55 $\pm$ 3.82	9.19 $\pm$ 10.52	9.28 $\pm$ 12.36
	Clinical	10.44 $\pm$ 2.25	6.61 $\pm$ 3.80	6.70 $\pm$ 4.34
	Biospecimen	36.57 $\pm$ 1.90	40.17 $\pm$ 9.75	38.81 $\pm$ 5.94
ABCD	Imaging	19.21 $\pm$ 9.62	14.04 $\pm$ 23.72	9.98 $\pm$ 18.99
	Genetics	30.90 $\pm$ 1.04	57.05 $\pm$ 14.71	28.75 $\pm$ 2.24
	Cog./health	23.88 $\pm$ 6.43	12.79 $\pm$ 7.73	5.95 $\pm$ 2.75
	Behav./env.	26.01 $\pm$ 3.09	16.12 $\pm$ 2.98	8.42 $\pm$ 2.22

Table 7 and Fig. 2 show that label-free prediction changes recover the main ordering of label-dependent performance effects. Pearson correlations between normalized change and accuracy-drop shares are 0.979 on ADNI and 0.825 on ABCD; both datasets have Spearman $\rho = 0.8$. ADNI concentrates functional dependence in structural MRI and biospecimens, consistent with its neurodegenerative endpoint. In ABCD, direct QC/LD-pruned SNP dosage has the largest prediction-change share and causes a 28.75-point mean accuracy decrease when strictly removed. Imaging, cognition–health, and behavior–environment retain secondary contributions, but none substitutes for the genetic signal. This ordering is obtained from interventions on the frozen multimodal classifier rather than from marginal feature distributions or an external PRS.

5. Discussion

Coupling completion with controlled trust. CERD leads all three main metrics on ADNI and ABCD. The incomplete-stratum control shows where reconstruction matters: conditional tokens recover decisions that are unavailable to an observed-subset-only model, while provenance and modality reliability regulate how that recovered evidence enters the branch mixture. This coupling prevents completion from becoming an isolated representation objective: its output is

evaluated according to both origin and task-dependent trust before contributing to the final probability.

Sparse specialization of completed evidence. Provenance exposes content, modality position, and evidence origin before expert routing. The one-expert controls lower accuracy and macro-F1 in both cohorts, and the capacity-aligned ABCD Dense FFN loses most clearly on incomplete participants. Since the Dense control retains the encoders, completion, token layout, and readouts, this gap isolates content-dependent expert transformations under changing source availability rather than capacity alone.

Why multigranular decomposition matters. Reducing CERD to its global head produces the largest accuracy loss in both datasets. Since each $\mathbf{f}_n^m$ already contains cross-modal context, modality and pair anchors expose source-centered agreement–discrepancy views beyond one pooled representation without treating them as independent unimodal predictions. Their normalized mixture also keeps these views auditable in the same probability that is used for classification.

Prediction consistency. ADNI depends most on structural MRI and biospecimens, whereas the severity-defined ABCD decision is most sensitive to direct SNP dosage. Prediction-change shares track independently measured accuracy-drop shares in both cohorts, connecting the model’s label-free response to a source intervention with its measured predictive dependence. The cohort-specific ordering is not imposed by the fusion rule; it emerges from the same architecture trained on distinct endpoints and source groups.

6. Conclusion

CERD unifies conditional token recovery, provenance-aware expert fusion, and reliability-based evidence decomposition. It ranks first for all three reported metrics on ADNI and family-disjoint ABCD severity classification. Matched controls isolate completion, sparse specialization, and multigranular readout, while strict-removal agreement connects output sensitivity to measured source dependence.

References

- [1] Baltrusaitis, T., Ahuja, C., Morency, L.P., 2019. Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 41, 423–443. doi:10.1109/TPAMI.2018.2798607.
- [2] Borys, K., Schmitt, Y.A., Nauta, M., Seifert, C., Krämer, N., Friedrich, C.M., Nensa, F., 2023. Explainable ai in medical imaging: An overview for clinical practitioners - saliency-based xai approaches. *European Journal of Radiology*.
- [3] Dorent, R., Joutard, S., Modat, M., Ourselin, S., Vercauteren, T., 2019. Hetero-modal variational encoder-decoder for joint modality completion and segmentation, in: *Medical Image Computing and Computer Assisted Intervention – MICCAI 2019*, Springer. pp. 74–82. doi:10.1007/978-3-030-32245-8_9.
- [4] Esteva, A., Robicquet, A., Ramsundar, B., Kuleshov, V., DePristo, M., Chou, K., Cui, C., Corrado, G., Thrun, S., Dean, J., 2019. A guide to deep learning in healthcare. *Nature Medicine*.
- [5] Fedus, W., Zoph, B., Shazeer, N., 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research* 23, 1–39.
- [6] Feng, Y., Gao, B., Deng, S., Qiu, A., Qin, J., 2024. Unified multi-modal learning for any modality combinations in alzheimer’s disease diagnosis, in: *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*, Springer Nature Switzerland. pp. 487–497. doi:10.1007/978-3-031-72384-1_46.
- [7] Gale, T., Narayanan, D., Young, C., Zaharia, M., 2023. MegaBlocks: Efficient sparse training with mixture-of-experts. *Proceedings of Machine Learning and Systems* 5.
- [8] Garavan, H., Bartsch, H., Conway, K., Decastro, A., Goldstein, R.Z., Heeringa, S., Jernigan, T., Potter, A., Thompson, W., Zahs, D., 2018. Recruiting the ABCD sample: Design considerations and procedures. *Developmental Cognitive Neuroscience* 32, 16–22. doi:10.1016/j.dcn.2018.04.004.
- [9] Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q., 2017. On calibration of modern neural networks, in: *Proceedings of the 34th International Conference on Machine Learning*, PMLR. pp. 1321–1330.
- [10] Guo, Z., Jin, T., Zhao, Z., 2024. Multimodal prompt learning with missing modalities for sentiment analysis and emotion recognition, in: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics. pp. 1726–1736. doi:10.18653/v1/2024.acl-long.94.
- [11] Havaei, M., Guizard, N., Larochelle, H., Jodoin, P., 2016. Hemis: Hetero-modal image segmentation. *Medical Image Computing and Computer-Assisted Intervention*.
- [12] Jack, C.R., Bennett, D.A., Blennow, K., Carrillo, M.C., Dunn, B., Haeberlein, S.B., Holtzman, D.M., Jagust, W., Jessen, F., Karlawish, J., et al., 2018. NIA-AA research framework: Toward a biological definition of alzheimer’s disease. *Alzheimer’s & Dementia*.
- [13] Jin, P., Zhu, B., Yuan, L., Yan, S., 2025. MoE++: Accelerating mixture-of-experts methods with zero-computation experts, in: *International Conference on Learning Representations*.
- [14] Lakshminarayanan, B., Pritzel, A., Blundell, C., 2017. Simple and scalable predictive uncertainty estimation using deep ensembles, in: *Advances in Neural Information Processing Systems*.
- [15] Lane, C.A., Hardy, J., Schott, J.M., 2018. Alzheimer’s disease. *European Journal of Neurology*.
- [16] Lee, Y.L., Tsai, Y.H., Chiu, W.C., Lee, C.Y., 2023. Multimodal prompting with missing modalities for visual recognition, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 14943–14952. doi:10.1109/CVPR52729.2023.01435.
- [17] Lepikhin, D., Lee, H., Xu, Y., Chen, D., Firat, O., Huang, Y., Krikun, M., Shazeer, N., Chen, Z., 2021. GShard: Scaling giant models with conditional computation and automatic sharding, in: *International Conference on Learning Representations*.
- [18] Lewis, M., Bhosale, S., Dettmers, T., Goyal, N., Zettlemoyer, L., 2021. BASE layers: Simplifying training of large, sparse models, in: *Proceedings of the 38th International Conference on Machine Learning*, PMLR. pp. 6265–6274.
- [19] Li, S., Chen, C., Han, J., 2025a. SimMLM: A simple framework for multi-modal learning with missing modality, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 24068–24077.
- [20] Li, Y., Dong, Y., Wu, Y., Yan, H., Gao, L., 2025b. Alzheimer’s disease recognition based on adaptive graph normalization flow for incomplete multimodal data fusion, in: *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*, Springer Nature Switzerland. pp. 64–73. doi:10.1007/978-3-032-04984-1_7.
- [21] Lin, R., Hu, H., 2023. MissModal: Increasing robustness to missing modality in multimodal sentiment analysis. *Transactions of the Association for Computational Linguistics* 11, 1686–1702. doi:10.1162/tac1_a_00628.
- [22] Liu, M., Cheng, D., Yan, W., Alzheimer’s Disease Neuroimaging Initiative, 2018. Classification of alzheimer’s disease by combination of convolutional and recurrent neural networks using fdg-pet images. *Frontiers in Neuroinformatics*.
- [23] Lundberg, S.M., Lee, S., 2017. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*.
- [24] Ma, M., Ren, J., Zhao, L., Tulyakov, S., Wu, C., Peng, X., 2021. Smil: Multimodal learning with severely missing modality. *Proceedings of the AAAI Conference on Artificial Intelligence*.
- [25] Mustafa, B., Riquelme, C., Puigcerver, J., Jenatton, R., Houlsby, N., 2022. Multimodal contrastive learning with LIMoE: The language–image mixture of experts, in: *Advances in Neural Information Processing Systems*. doi:10.52202/068431-0695.
- [26] Ngiam, J., Khosla, A., Kim, M., Nam, J., Lee, H., Ng, A.Y., 2011. Multimodal deep learning, in: *Proceedings of the 28th International Conference on Machine Learning*, pp. 689–696.
- [27] Ovadia, Y., Fertig, E., Ren, J., Nado, Z., Sculley, D., Nowozin, S., Dillon, J., Lakshminarayanan, B., Snoek, J., 2019. Can you trust your model’s uncertainty? evaluating predictive uncertainty under dataset shift, in: *Advances in Neural Information Processing Systems*.
- [28] Parcalabescu, L., Frank, A., 2023. MM-SHAP: A performance-agnostic metric for measuring multimodal contributions in vision and language models and tasks, in: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics. pp. 4032–4059. doi:10.18653/v1/2023.acl-long.223.
- [29] Park, J., Park, K., Kim, D., Park, J., Choi, J.W., 2025. Layer-wise modality decomposition for interpretable multimodal sensor fusion, in: *Advances in Neural Information Processing Systems*.
- [30] Rahman, W., Hasan, M.K., Lee, S., Zadeh, A.B., Mao, C., Morency, L.P., Hoque, E., 2020. Integrating multimodal

- information in large pretrained transformers. Proceedings of the 58th annual meeting of the association for computational linguistics .
- [31] Ribeiro, M.T., Singh, S., Guestrin, C., 2016. “why should i trust you?”: Explaining the predictions of any classifier. Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining .
- [32] Riquelme, C., Puigcerver, J., Mustafa, B., Neumann, M., Jenatton, R., Pinto, A.S., Keysers, D., Houlsby, N., 2021. Scaling vision with sparse mixture of experts, in: Advances in Neural Information Processing Systems.
- [33] Rudin, C., 2019. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence* .
- [34] Sadeghi, Z., Alizadehsani, R., Cifci, M.A., Kausar, S., Rehman, R., Mahanta, P., Bora, P.K., Almasri, A., Alkhawaldeh, R.S., Hussain, S., et al., 2024. A review of explainable artificial intelligence in healthcare. *Computers and Electrical Engineering* .
- [35] Scheltens, P., De Strooper, B., Kivipelto, M., Holstege, H., Chételat, G., Teunissen, C.E., Cummings, J., van der Flier, W.M., 2021. Alzheimer’s disease. *The Lancet* .
- [36] Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D., 2017. Grad-cam: Visual explanations from deep networks via gradient-based localization. Proceedings of the IEEE international conference on computer vision .
- [37] Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., Dean, J., 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, in: International Conference on Learning Representations.
- [38] Suk, H.I., Lee, S.W., Shen, D., Alzheimer’s Disease Neuroimaging Initiative, 2014. Hierarchical feature representation and multimodal fusion with deep learning for ad/mci diagnosis. *NeuroImage* .
- [39] Sundararajan, M., Taly, A., Yan, Q., 2017. Axiomatic attribution for deep networks. International Conference on Machine Learning .
- [40] Sutter, T.M., Daunhawer, I., Vogt, J.E., 2021. Generalized multimodal elbo. International Conference on Learning Representations .
- [41] Tonekaboni, S., Joshi, S., McCradden, M.D., Goldenberg, A., 2019. What clinicians want: contextualizing explainable machine learning for clinical end use. Machine learning for healthcare conference .
- [42] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I., 2017. Attention is all you need, in: Advances in Neural Information Processing Systems.
- [43] Venugopalan, J., Tong, L., Hassanzadeh, H.R., Wang, M.D., 2021. Multimodal deep learning models for early detection of alzheimer’s disease stage. *Scientific Reports* .
- [44] Wachter, S., Mittelstadt, B., Russell, C., 2018. Counterfactual explanations without opening the black box. *Harvard Journal of Law & Technology* .
- [45] Wang, H., Chen, Y., Ma, C., Avery, J., Hull, L., Carneiro, G., 2023a. Multi-modal learning with missing modality via shared-specific feature modelling, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 15878–15887.
- [46] Wang, Y., Cui, Z., Li, Y., 2023b. Distribution-consistent modal recovering for incomplete multimodal learning, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 22025–22034.
- [47] Weiner, M.W., Veitch, D.P., Aisen, P.S., Beckett, L.A., Cairns, N.J., Green, R.C., Harvey, D., Jack, C.R., Jagust, W., Liu, E., et al., 2013. The alzheimer’s disease neuroimaging initiative: A review of papers published since its inception. *Alzheimer’s & Dementia* .
- [48] Woo, S., Lee, S., Park, Y., Nugroho, M.A., Kim, C., 2023. Towards good practices for missing modality robust action recognition, in: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 2776–2784. doi:10.1609/aaai.v37i3.25378.
- [49] Wu, M., Goodman, N.D., 2018. Multimodal generative models for scalable weakly-supervised learning. *Advances in Neural Information Processing Systems* .
- [50] Wu, R., Wang, H., Chen, H.T., Carneiro, G., 2024. Deep multimodal learning with missing modality: A survey. arXiv preprint arXiv:2409.07825 .
- [51] Xin, J., Yun, S., Peng, J., Choi, I., Ballard, J.L., Chen, T., Long, Q., 2025. I^2 MoE: Interpretable multimodal interaction-aware mixture-of-experts, in: Proceedings of the 42nd International Conference on Machine Learning, PMLR. pp. 68870–68888.
- [52] Yang, Z., Wang, H., Hu, D., 2025. Efficient quantification of multimodal interaction at sample level, in: Proceedings of the 42nd International Conference on Machine Learning, PMLR. pp. 71302–71317.
- [53] Yu, H., Qi, Z., Jang, L.K., Salakhutdinov, R., Morency, L.P., Liang, P.P., 2024. MMoE: Enhancing multimodal models with mixtures of multimodal interaction experts, in: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics. pp. 10006–10030. doi:10.18653/v1/2024.emnlp-main.558.
- [54] Yun, S., Choi, I., Peng, J., Wu, Y., Bao, J., Zhang, Q., Xin, J., Long, Q., Chen, T., 2024. Flex-moe: Modeling arbitrary modality combination via the flexible mixture-of-experts. *Advances in Neural Information Processing Systems* .
- [55] Zhan, Y., Yang, R., You, J., Huang, M., Liu, W., Liu, X., 2025. A systematic literature review on incomplete multimodal learning: techniques and challenges. *Systems Science & Control Engineering* .
- [56] Zhang, C., Chu, X., Ma, L., Zhu, Y., Wang, Y., Wang, J., Zhao, J., 2022. M3care: Learning with missing modalities in multimodal healthcare data. Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining .
- [57] Zhang, D., Wang, Y., Zhou, L., Yuan, H., Shen, D., Initiative, A.D.N., et al., 2011. Multimodal classification of alzheimer’s disease and mild cognitive impairment. *NeuroImage* .
- [58] Zhao, J., Li, R., Jin, Q., 2021. Missing modality imagination network for emotion recognition with uncertain missing modalities, in: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing, Association for Computational Linguistics. pp. 2608–2618. doi:10.18653/v1/2021.acl-long.203.
- [59] Zhou, R., Zhou, H., Su, Y., Chen, B.Y., Zhang, Y., He, L., Alzheimer’s Disease Neuroimaging Initiative, 2026. Adaptive clinical-aware latent diffusion for multimodal brain image generation and missing modality imputation. arXiv preprint arXiv:2603.09931 doi:10.48550/arXiv.2603.09931.
- [60] Zhou, Y., Lei, T., Liu, H., Du, N., Huang, Y., Zhao, V., Dai, A.M., Chen, Z., Le, Q.V., Laudon, J., 2022. Mixture-of-experts with expert choice routing, in: Advances in Neural Information Processing Systems. doi:10.52202/068431-0515.